

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Regular/Supplementary Winter Examination – 2024****Course:** Design of Rein. & Prestressed Conc. Struct.**Branch:** Civil Engineering**Semester:** VII**Subject Code & Name:** BTCVC 701**Max Marks:** 60**Date:** 05/02/2025**Duration:** 3 Hr.**Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.
7. Use of IS 456 and IS 1343 is permitted.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	For a singly reinforced concrete beam $b = 300$ mm, $d = 600$ mm, $D = 650$ mm, subjected to factored $M_u = 180$ kNm, factored $T_u = 75$ kNm and factored $V_u = 90$ kN. The equivalent shear for which beam need to design is: a. 350 kN b. 390 kN c. 490 kN d. 590 kN	L3/CO1	1
2	Find the equivalent shear stress in RC beam having $b = 230$ mm, $d = 460$ mm and equivalent shear $V_e = 175$ kN. a. 1.55 MPa b. 1.65 MPa c. 2.65 MPa d. 2.43 MPa	L3/CO1	1
3	For a RC beam 300 mm x 600 mm, grade of concrete M25 and 0.875 % of tension steel, design shear strength of concrete is: a. 0.60 MPa b. 0.62 MPa c. 0.64 MPa d. 0.66 MPa	L3/CO1	1
4	The minimum eccentricity in a column should not exceed.... of lateral dimension. a. 0.04 b. 0.05 c. 0.06 d. 0.07	L2/CO2	1
5	For square column with cross section 300 mm x 300 mm having effective length of 2500 mm, what is the minimum eccentricity as per IS 456? a. 15 mm b. 20 mm c. 30 mm d. 40 mm	L2/CO2	1
6	If the diameter of longitudinal bars in a square column is 20 mm, the diameter of lateral ties should not be less than: a. 4 mm b. 5 mm c. 6 mm d. 8 mm	L2/CO2	1
7	For design of column members subjected to combined axial load and biaxial bending, the ratio of moments about x and y axis should be:	L2/CO2	1

	a. Equal to 1	b. Greater than 1	c. Less than 1	d. Not related to ratio		
8	Railway sleepers are example of:				L1/CO3	1
	a. Post tension member	b. Pre tension member	c. RCC member	d. Bending member		
9	Prestressed concrete is widely used in construction of:				L1/CO3	1
	a. Bore well	b. Houses	c. Nuclear power vessels	d. Roofs		
10	Loss of prestress with time at constant strain in steel is called as:				L1/CO3	1
	a. _____	b. _____	c. _____	d. _____		
11	Which of the following influence the deflections of prestressed concrete members?				L2/CO4	1
	a. Wall Profile	b. Types of Aggregates	c. Type of cement	d. Cable profile		
12	In the anchorage zone or the end block of a post tensioned prestressed concrete element, the state of stress distribution is considered as:				L1/CO4	1
	a. Unity	b. Zero	c. Complex	d. Easy		
Q.2	Solve the following.					12
A)	A reinforced concrete rectangular beam $b = 300$ mm, $d = 500$ mm and $D = 550$ mm is subjected to factored shear force $V_u = 75$ kN, $M_u = 170$ kNm, $T_u = 50$ kNm. Assume M 20 concrete, Fe 415 steel. Design the beam for limit state of collapse in torsion.				L3/CO1	6
B)	A reinforced concrete rectangular beam of $b = 300$ mm, $d = 600$ mm and $D = 650$ mm subjected to factored shear force of 90 kN. Assuming % age of steel as 0.75 in the section, determine the factored torsional moment in the section if a) No additional reinforcement for torsion is provided b) Maximum steel for torsion is provided in the section. Assume M 25 Grade concrete and Fe 415 Steel.				L3/CO1	6
Q.3	Solve the following.					12
A)	Find the load carrying capacity of RC column 230×450 mm subjected to axial load. The column has unsupported length of 3.0 m and effectively held in position and restrained at one end only. The column is reinforced with 20 mm dia. 6 bars at corners. Use M25 grade concrete and Fe 415 steel.				L3/CO2	6
B)	Design a short spiral column subjected to $P_u = 1500$ kN and $M_u = 150$ kNm using M 25 and Fe 415. The preliminary diameter of the column may be taken as 500 mm. Consider effective cover as 50 mm. Use following table: p- percentage of steel.				L3/CO2	6

		Mu/fck D ³	Pu/fckD ²	p/fck			
		0.04	0.2	0.01			
		0.05	0.24	0.03			
Q. 4	Solve Any Two of the following.						12
A)	Differentiate between pre-tensioning and post-tensioning methods of prestressed concrete.					L2/CO3	6
B)	Draw and explain different cable profile used in prestressed beam with their suitability.					L2/CO3	6
C)	Explain Magnel-Blaton system of prestressing with neat sketch.					L2/CO3	6
Q.5	Solve Any Two of the following.						12
A)	List and explain the types of losses in the prestressed (pre-tensioned and post-tensioned) concrete beam.					L2/CO3	6
B)	A pre-tensioned concrete beam has a cross section 250 x 360 mm prestressed by cable having a cross sectional area of 504 mm ² . The centroid of the cable is located at 105 mm from the soffit. Find the percentage loss of stress in steel due to creep, shrinkage, elastic deformation of concrete and 5% relaxation of stress in steel. The initial pre-stress in steel is 1000 MPa, E _s = 210 GPa, E _c = 36.8 GPa, Creep coefficient = 1.6, Shrinkage strain = 1.9 x 10 ⁻⁴ .					L3/CO3	6
C)	A rectangular concrete beam 200 mm wide and 450 mm deep spanning over 8 m is prestressed by a straight cable carrying an effective prestressing force of 500 kN located at an eccentricity of 50 mm. The beam supports a live load of 4 kN/m. Calculate the resultant stress distribution for the center of span of the beam assuming the density of concrete as 25 kN/m ³ .					L3/CO3	6
Q. 6	Solve Any Two of the following.						12
A)	What is structural auditing? What are its advantages and how its carried? Explain with example.					L2/CO4	6
B)	A prestressed concrete beam of rectangular section 300 mm wide and 600 mm deep has a span of 15 m. The effective force is 1520 kN at an eccentricity of 150 mm. The DL of the beam is 5 kN/m and LL on beam is 8 kN/m. Determine the extreme stresses in concrete at i) end span, ii) at mid span.					L3/CO4	6
C)	Design a post-tensioned hanger to carry an axial tension of PDL = 225 kN (dead load including self-weight) and PLL = 125 kN. The dimension of the hanger is 300 × 300 mm. Design the section without considering non-prestressed reinforcement. Tension is not allowed under service loads. The grade of concrete is M 35. The age at transfer is 28 days. Assume 15% long term losses in the prestress. The					L3/CO4	6

	following properties of the prestressing strands are available from tests. Type of prestressing tendon : 7 wire strand, nominal diameter = 12.8 mm, nominal area = 99.3 mm ² , tensile strength (f _{pk}) = 1860 N/mm ² , Modulus of elasticity = 195 kN/mm ² .		
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